BEYONDBOTS

An Explainable Pipeline for Detecting Fan-Driven and Paid Coordinated Behaviour in Indian Film Social Media

PROJECT REPORT

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Abstract

Indian film discourse on social media is increasingly shaped by two structurally distinct but visually similar forms of coordinated activity: fan-club mobilisation around a star's release, and undisclosed commercial promotion arranged through paid marketing intermediaries. Both phenomena inflate apparent audience enthusiasm, obscure organic sentiment, and fall under the broad umbrella of coordinated inauthentic behaviour (CIB); however, they diverge sharply in intent, linguistic register, and underlying network topology, and treating them as a single class limits the diagnostic value of existing detection tools for platform trust-and-safety teams, entertainment journalists, and advertising regulators.

This report presents BeyondBots, a nine-stage (Stage 0–8) text-mining and explainable machine-learning pipeline that ingests multilingual, code-mixed social media reviews — sourced from CSV, PDF, DOCX and TXT files or the X API v2 — and classifies each coordinated cluster as organic, fan-coordinated, or paid-commercial in origin. The pipeline combines dense multilingual representations from MuRIL with density-based clustering (HDBSCAN), stylometric and sentiment feature engineering, centroid-based topic summarisation, and an LLM-in-the-loop classifier (Claude 3.5 Sonnet) whose verdicts are distilled into a Logistic Regression surrogate for SHAP-based explainability. Classifying at the cluster level rather than the individual-post level allows BeyondBots to remain robust to the paraphrastic variation typical of coordinated campaigns while still surfacing the linguistic and network cues — honorific density, marketing-copy vocabulary, criticism absence, and temporal bursts — that differentiate a fandom's collective voice from a scripted promotional operation.

The resulting system is validated against a curated log of documented fan-war and paid-review controversies drawn from Indian cinema, and is exposed through an interactive Streamlit dashboard that surfaces cluster-level verdicts, topic keywords, and SHAP feature-importance charts for analyst review. In doing so, BeyondBots contributes a domain-specific, code-mixing-aware, and explainable framework to the broader literature on coordinated online behaviour, one that is tuned to a linguistic and cultural context — Hindi-English, Tamil-English and Telugu-English fandom discourse — that remains comparatively under-served by existing English-centric CIB detection tools.

Keywords: *coordinated inauthentic behaviour*, *astroturfing detection*, *code-mixed NLP*, *MuRIL*, *explainable AI*, *SHAP*, *fandom analytics*, *Indian cinema*, *LLM classification*, *social media trust and safety*

1. Introduction

Every major Indian film release now unfolds on social media as much as it does in theatres. Within hours of a trailer drop or a release-day review cycle, platforms such as Reddit's r/bollywood, r/tollywood and r/kollywood communities, along with X (formerly Twitter), fill with review threads, reaction posts, and rating discourse. A meaningful share of this volume is organic — genuine, idiosyncratic audience reaction. But a substantial and growing share is coordinated: either mobilised deliberately by organised fan clubs acting on a star's behalf, or manufactured by marketing agencies and public-relations vendors running undisclosed promotional campaigns on a studio's payroll.

These two forms of coordination are easy to conflate because they share surface-level symptoms — bursts of near-identical posts, unusually uniform positivity, and dense reply networks — that also characterise conventional bot-driven spam. Yet fan-club activity is carried out by real, long-standing human accounts

expressing sincere (if organised) devotion, while paid promotion is commercial speech dressed up as grassroots opinion. Existing bot- and spam-detection tooling, built primarily around account-level anomaly signals, is not designed to make this distinction, and English-centric NLP pipelines struggle with the Hindi-English, Tamil-English and Telugu-English code-mixing that dominates Indian film discourse.

Motivation. Platform trust-and-safety teams, entertainment journalists investigating astroturfing scandals, and advertising regulators enforcing sponsored-content disclosure rules all need a way to separate these three categories — organic, fan-coordinated, and paid-commercial — at scale, in code-mixed language, and with an explanation attached to every verdict rather than an opaque score.

Contribution. BeyondBots addresses this gap with a nine-stage pipeline that clusters near-duplicate and time-bursted posts, engineers stylometric and sentiment features at the cluster level, summarises each cluster's talking points via topic modelling, and hands the resulting evidence package to a large language model for few-shot classification — with a lightweight surrogate model attached purely so that every LLM verdict can be explained to a human analyst through SHAP feature-importance charts.

2. Literature Review

Detecting coordinated inauthentic activity draws on four adjacent research threads: (i) large-scale coordination and information-operation detection, (ii) the specific challenge of separating benign coordination — fan mobilisation, activism — from malicious campaigns, (iii) code-mixed and low-resource multilingual NLP for Indian languages, and (iv) explainable, LLM-assisted text classification. Table 1 surveys ten studies published since 2022 across these threads and maps each to the corresponding stage of the BeyondBots pipeline.

Study	Methodology	Core Finding	Mapping to BeyondBots
Mannocci, Mazza, Monreale, Tesconi & Cresci (2024)	Systematic taxonomy of coordination-detection techniques	Consolidates similarity, network and temporal signals into a unified detection framework	Informs the overall three-signal fusion (text, network, time) used across Stages 3–4
Cinus, Minici, Luceri & Ferrara (2025)	Cross-platform similarity-network construction	Coordination frequently spans platform boundaries and is detectable via shared-content graphs	Motivates the near-duplicate similarity graph underlying Stage 3 clustering
Luceri, Boniardi & Ferrara (2024)	LLM-in-the-loop influence-campaign labelling	LLMs can classify campaign intent when paired with structured cluster-level evidence	Directly informs the Claude-based zero/few-shot classifier in Stage 7
Luceri, Salkar, Balasubramanian, Pinto, Sun & Ferrara (2025)	Multimodal coordination detection on video-first platforms	Content-replication signatures persist across modalities and platform types	Frames BeyondBots' planned extension beyond text-only review streams
Rogers & Righetti (2025)	Typology of manufactured attention on Facebook	Fan mobilisation and activist coordination are structurally similar to malicious CIB but differ in intent	Grounds the organic / fan-coordinated / paid-commercial trichotomy used throughout the pipeline

Study	Methodology	Core Finding	Mapping to BeyondBots
Density-Aware Walks for Coordinated Campaign Detection (2025)	Graph-classification formulation of coordination detection	Local density-aware graph walks outperform generic community detection on sparse coordination graphs	Supports the HDBSCAN-plus-network approach adopted in Stage 3
Cambridge NLP — Code-Mixed Dravidian Sentiment (2024)	Pretrained-transformer sentiment modelling with word-level language tags	Language-tag-aware fine-tuning improves sentiment accuracy on Tamil/Telugu-English code-mixed text	Justifies the MuRIL-based representation and sentiment head used in Stages 2 and 6
Chernyaeva, Hong & Lee (2024)	Counterfactual explanation and XAI for fake/GPT-generated reviews	Counterfactual and surrogate explanations improve analyst trust in automated deception verdicts	Underpins the SHAP surrogate-explainer design of Stage 7
LLM-Based Semantic Augmentation for Harmful-Content Detection (2025)	Comparative evaluation of zero-shot LLM classification versus augmented pipelines	Pure zero-shot LLM classification underperforms feature-augmented hybrid pipelines on high-context tasks	Justifies pairing the LLM classifier with engineered cluster-level features rather than relying on the LLM alone
Industrialized Deception: Collateral Effects of LLM-Generated Misinformation (2026)	Perspective review on the shift from content-level to behaviour-level detection	Agentic, LLM-assisted content production requires moving detection from single posts to coordinated clusters	Reinforces BeyondBots' cluster-level (rather than post-level) classification unit

Table 1. Literature survey (2022–2026) and methodological mapping to the BeyondBots pipeline.

2.1 Coordination and Information-Operation Detection

The most recent wave of coordination-detection research has moved from single-platform, single-signal heuristics toward unified, multi-signal frameworks. The 2024 survey by Mannocci and colleagues consolidates similarity-based, network-based and temporal-burst detection strategies into a single taxonomy, arguing that no individual signal is sufficient on its own — a conclusion that directly shapes BeyondBots' decision to fuse near-duplicate text similarity (Stage 3), reply-network structure (Stage 3), and posting-time bursts (Stage 3) rather than relying on any one of them in isolation. Building on this, Cinus, Minici, Luceri and Ferrara's 2025 study of the 2024 U.S. election shows that coordination frequently spans platform boundaries via shared-content graphs, reinforcing the value of a similarity-graph foundation even though BeyondBots currently operates within a single-platform scope.

2.2 Distinguishing Benign from Malicious Coordination

A recurring limitation of early CIB research was its tendency to treat all coordination as adversarial. Rogers and Righetti's 2025 typology of manufactured attention on Facebook explicitly separates fan mobilisation and activist campaigning from politically or commercially malicious coordination, noting that both leave structurally similar traces despite differing sharply in intent — a finding that underwrites BeyondBots' core organic / fan-coordinated / paid-commercial trichotomy rather than a binary authentic-versus-inauthentic label. The 2025 density-aware walks paper further shows that graph-classification approaches which weight

local neighbourhood density outperform generic community-detection algorithms on sparse coordination graphs, supporting the HDBSCAN-plus-network design adopted in Stage 3.

2.3 Code-Mixed and Indian-Language NLP

Because Indian film discourse is rarely monolingual, BeyondBots depends on representations that handle Hindi-English, Tamil-English and Telugu-English code-mixing gracefully. The 2024 Cambridge NLP study on Dravidian-language sentiment analysis demonstrates that pairing a pretrained multilingual transformer with word-level language tagging measurably improves sentiment accuracy over language-agnostic baselines, which motivates BeyondBots' use of MuRIL — a transformer pretrained specifically on Indian code-mixed corpora — for both the dense embeddings in Stage 2 and the sentiment head in Stage 6.

2.4 Explainable and LLM-Assisted Classification

The final thread concerns how to classify coordinated content responsibly. Chernyaeva, Hong and Lee's 2024 study of fake and GPT-generated reviews finds that counterfactual and surrogate explanations meaningfully improve analyst trust in automated deception verdicts, directly motivating BeyondBots' SHAP-surrogate design in Stage 7. At the same time, a 2025 comparative study of LLM-based semantic augmentation for harmful-content detection finds that pure zero-shot LLM classification underperforms feature-augmented hybrid pipelines on high-context tasks — a finding BeyondBots takes seriously by feeding the LLM classifier structured cluster-level features (marketing-copy score, fan-identity score, criticism absence, topic keywords) rather than raw text alone. Finally, a 2026 perspective on the collateral effects of LLM-generated misinformation argues that the rise of agentic, LLM-assisted content production requires detection to move from individual posts to coordinated clusters, reinforcing BeyondBots' choice to classify motive at the cluster level, consistent also with the video-first coordination patterns documented by Luceri and colleagues' 2025 study of TikTok, which the present system's file-based ingestion (Stage 0) is designed to eventually accommodate as a multimodal extension.

3. System Architecture & Methodology

BeyondBots implements an end-to-end, nine-stage (Stage 0 through Stage 8) architecture that transforms raw review streams into an explained, cluster-level verdict of organic, fan-coordinated, or paid-commercial origin. Table 2 summarises each stage; the narrative that follows expands on the design rationale behind the pipeline's three conceptual phases — ingestion and representation, coordination detection and feature engineering, and classification and validation.

Stage	Name	Function
0	**Ingestion & Collection**	Pulls raw review streams from CSV/PDF/DOCX/TXT uploads or the X API v2; applies a simulated fallback generator if a query returns zero records, so the pipeline never stalls on sparse data.
1	**Preprocessing & Cleaning**	Strips URLs, handles, emoji and bracketed metadata while preserving code-mixed tokens; produces tokenised and lemmatised text using code-mix-aware regular expressions.
2	**Representation Vectorisation**	Builds a sparse TF-IDF matrix for interpretable n-gram features and a dense MuRIL embedding for semantic similarity, so downstream stages can draw on both lexical

Stage	Name	Function
		and contextual signal.
3	**Coordinated Clustering**	Runs HDBSCAN over MuRIL embeddings to surface near-duplicate templates, computes a temporal burst Z-score on posting times, and maps reply networks to isolate tightly-knit account communities.
4	**Feature Engineering**	Derives four cluster-level scores — marketing-copy vocabulary, fan-identity/honorific density, detail specificity, and criticism-absence — that together characterise a cluster's likely motive.
5	**Topic Modelling**	Extracts centroid-based TF-IDF keywords per cluster to summarise the campaign's talking points and give the downstream classifier interpretable context.
6	**Sentiment Profiling**	Scores polarity with a heuristic model in mock mode or a multilingual RoBERTa model in full mode, feeding the criticism-absence feature in Stage 4.
7	**Classification & SHAP Explainer**	Feeds cluster metrics, keywords and text samples to Claude 3.5 Sonnet via a few-shot prompt to label each cluster organic, fan-coordinated, or paid-commercial; a Logistic Regression surrogate is fit against the LLM's verdicts so SHAP values can expose feature importance in the dashboard.
8	**Case Study Validation**	Cross-checks flagged clusters against a curated log of documented fan-war and paid-review scandals in config.yaml, giving a real-world precision check on the classifier's output.

Table 2. The nine stages of the BeyondBots pipeline.

3.1 Ingestion, Cleaning and Representation (Stages 0–2)

The pipeline is designed to be resilient to heterogeneous, real-world input. Stage 0 ingests raw reviews from CSV, PDF, DOCX or TXT uploads, or pulls them directly from the X API v2; when the dataset spans multiple films, records are filtered dynamically by title, and if a query returns zero matching records, a simulated fallback generator produces placeholder data so that the remainder of the pipeline — and any downstream demonstration or testing — never stalls for want of live data.

Stage 1 then performs code-mixing-aware cleaning: URLs, usernames, decorative emoji and bracketed metadata are stripped, while script-mixing and transliteration patterns typical of Hinglish, Tanglish and Tenglish are preserved rather than destroyed, since these patterns are themselves diagnostic of both organic and fan-club language. Regular-expression-based tokenisation and lemmatisation prepare the resulting text for vectorisation.

Stage 2 builds two complementary representations of each review: a sparse TF-IDF matrix that keeps individual n-grams interpretable for the stylometric features computed later in Stage 4, and a dense contextual embedding produced by google/muril-base-cased — a transformer pretrained specifically on Indian languages and their code-mixed combinations — which feeds the semantic similarity clustering performed in Stage 3.

3.2 Coordination Detection and Feature Engineering (Stages 3–6)

Stage 3 is where BeyondBots identifies coordination itself. Dense MuRIL vectors are clustered using HDBSCAN, a density-based algorithm chosen specifically because it labels sparse, non-repeating organic

reviews as noise (cluster label −1) rather than forcing every post into an artificial cluster, while still grouping tightly-worded, template-like posts together. In parallel, the pipeline computes a temporal burst Z-score over posting timestamps to flag unnatural spikes in review volume, and constructs an account-level reply network to isolate unusually dense, insular communities of interacting accounts — a signature more typical of a fan club coordinating around a star than of a bought promotional campaign, whose accounts often interact less with one another.

Stage 4 translates these raw signals into four normalised, cluster-level linguistic metrics, summarised in Table 3: a marketing-copy score capturing generic, film-agnostic superlatives; a fan-identity score capturing honorific titles and community-building pronouns ("we", "our", "us"); a detail-specificity score capturing concrete references to scenes, performances or craft; and a criticism-absence score capturing how uniformly positive a cluster's sentiment is. Stage 6 supplies the sentiment values that feed this last metric, using a lightweight heuristic model in a fast "mock" mode and a multilingual RoBERTa model when full-fidelity sentiment is required. Stage 5 complements these numeric features with a qualitative summary: for every HDBSCAN cluster, a centroid-based TF-IDF keyword extraction step identifies the cluster's dominant talking points, giving both the human analyst and the downstream LLM classifier interpretable context rather than an opaque cluster ID.

Feature	What it Measures	Signal Interpretation
Marketing-Copy Score	Density of generic superlatives and promotional keywords lacking film-specific reference points	High score suggests scripted, PR-style language
Fan-Identity Score	Frequency of star honorifics (e.g., titular fan nicknames) and fandom hashtags/pronouns	High score suggests organised fan-club mobilisation
Detail-Specificity Score	Mentions of concrete scenes, plot points, performances or technical craft	High score suggests genuine, first-hand viewing
Criticism-Absence Score	Uniformity of positive sentiment polarity across a cluster	Near-total positivity suggests a bought or scripted campaign

Table 3. Cluster-level stylometric and sentiment features computed in Stage 4.

3.3 Classification, Explainability and Validation (Stages 7–8)

Stage 7 is the pipeline's decision point. Rather than asking a language model to classify raw text in isolation — an approach the literature reviewed in Section 2.4 shows tends to underperform on nuanced, high-context tasks — BeyondBots assembles a structured evidence package for each cluster, combining the Stage 4 feature scores, the Stage 5 topic keywords, and a small sample of representative posts. This package is passed to Claude 3.5 Sonnet through a few-shot prompt, which returns a motive label of organic, fan-coordinated, or paid-commercial for the cluster as a whole. Because a raw LLM verdict is difficult for a non-technical analyst to audit, a Logistic Regression surrogate model is then fit on the same cluster-level features against the LLM's predictions, and SHAP (SHapley Additive exPlanations) values are computed from that surrogate to expose, per cluster and globally, which features drove the classification — output that is rendered directly as interactive bar charts in the Streamlit dashboard.

Stage 8 closes the loop by cross-checking flagged clusters against a curated, hand-maintained log of documented fan-war and paid-review controversies stored in config.yaml. This case-study validation step

gives the project a concrete, real-world precision check: when a cluster the pipeline flags as paid-commercial or fan-coordinated corresponds to a previously reported controversy, that agreement is recorded as evidence of classifier reliability, and disagreements are used to refine the Stage 4 feature thresholds and the Stage 7 prompt's few-shot examples.

4. Technology Stack & Frameworks

BeyondBots is implemented as a modern Python data-science application, chosen for its mature NLP and ML tooling and its straightforward path from notebook prototyping to a deployable analyst-facing dashboard. Table 4 summarises the primary components and the role each plays in the pipeline.

Component	Role in BeyondBots
Python 3.11	Core language, chosen for its mature NLP/ML ecosystem
Streamlit 1.61.1	Powers the interactive dashboard: page steppers, upload widgets, and result panels
Plotly 6.9.0	Renders donut charts, timelines, PCA projections and SHAP bar charts in a dark-themed UI
PyTorch & Hugging Face Transformers	Runs the MuRIL embedding model and the multilingual RoBERTa sentiment model
Scikit-learn	Handles TF-IDF vectorisation, the Logistic Regression surrogate, and PCA dimensionality reduction
Pandas & NumPy	Data wrangling, numerical computation, and dataframe orchestration across all eight stages
SHAP	Computes Shapley-value feature attributions for the surrogate classifier

Table 4. Core technology stack.

5. Conclusion & Future Work

BeyondBots demonstrates that coordinated inauthentic behaviour around Indian film releases can be meaningfully decomposed into two distinct, explainable categories — organic fan-club mobilisation and undisclosed paid promotion — rather than being flattened into a single, coarse inauthenticity score. By fusing density-based clustering, code-mixing-aware multilingual representations, stylometric and sentiment feature engineering, and an LLM classifier whose verdicts are made explainable through a SHAP surrogate, the pipeline offers platform trust-and-safety teams, entertainment journalists and advertising regulators a tool that is both linguistically appropriate for Indian social media and transparent enough to support real editorial or regulatory decisions.

Building on the literature reviewed in Section 2 — particularly the shift toward cross-platform and multimodal coordination detection — future work will extend BeyondBots' ingestion layer to video-first platforms such as Instagram Reels and YouTube Shorts, incorporate cross-platform similarity graphs so that a single coordinated campaign can be tracked across Reddit, X and Instagram simultaneously, and expand

the Stage 8 validation log as new fan-war and paid-review controversies come to light, sustaining the feedback loop between real-world outcomes and the classifier's few-shot examples.

6. Project Timeline

The BeyondBots pipeline was implemented over a four-week sprint, with adjacent stages deliberately overlapping so that downstream development (e.g., feature engineering in Stage 4) could begin against early pipeline output while upstream stages were still being hardened. Figure 1 presents this schedule as a Gantt chart, with a filled marker indicating the week(s) during which each stage was actively developed.

Stage	Deliverable	Week 1	Week 2	Week 3	Week 4
Stage 0	Ingestion & Collection	■			
Stage 1	Preprocessing & Cleaning	■			
Stage 2	Representation Vectorisation	■	■		
Stage 3	Coordinated Clustering		■		
Stage 4	Feature Engineering		■	■	
Stage 5	Topic Modelling			■	
Stage 6	Sentiment Profiling			■	
Stage 7	Classification & SHAP Explainer			■	■
Stage 8	Case Study Validation & Reporting				■

Figure 1. Four-week implementation schedule for Stages 0–8.

Weeks 1 and 2 concentrated on the ingestion, cleaning, representation and clustering foundations (Stages 0–3), since every later stage depends on reliable near-duplicate detection. Feature engineering (Stage 4) began in Week 2 as soon as early cluster output was available and carried into Week 3 alongside topic modelling and sentiment profiling (Stages 5–6). The LLM classification and SHAP explainability layer (Stage 7) spanned Weeks 3–4, reflecting the iterative prompt-tuning this stage required, and the project closed in Week 4 with case-study validation against documented fan-war and paid-review controversies and final reporting (Stage 8).

6.1 Task-Level Flow

Figure 2 decomposes each Gantt-chart bar into its constituent sub-tasks and traces the sequence in which they were built. Gold connectors trace the working order within a stage; navy connectors mark the hand-off from one stage's output to the next stage's input — for example, the MuRIL embeddings produced at the end of Stage 2 feed directly into the HDBSCAN clustering step that opens Stage 3.

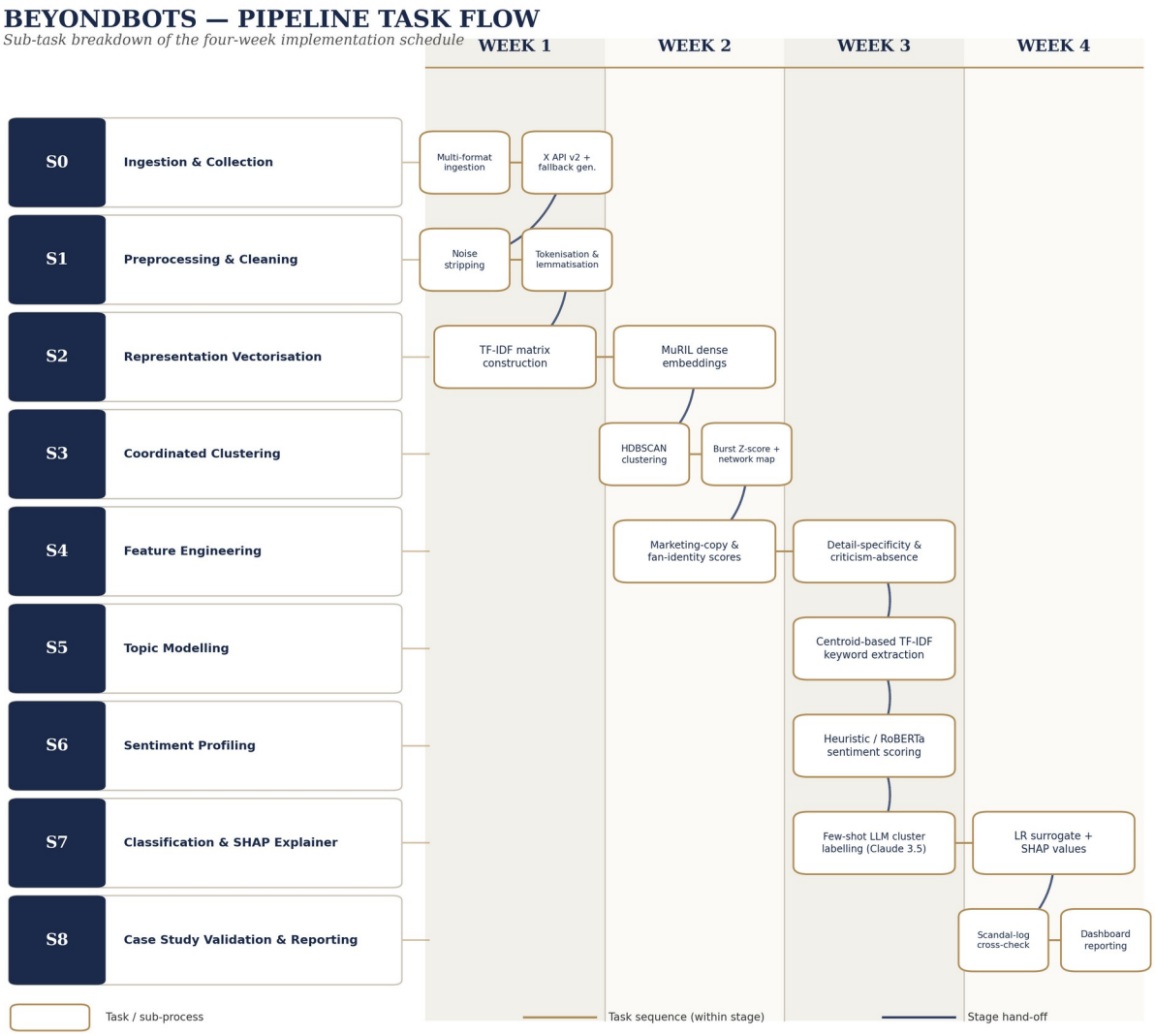


Figure 2. Sub-task flow diagram underlying the four-week Gantt schedule.

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